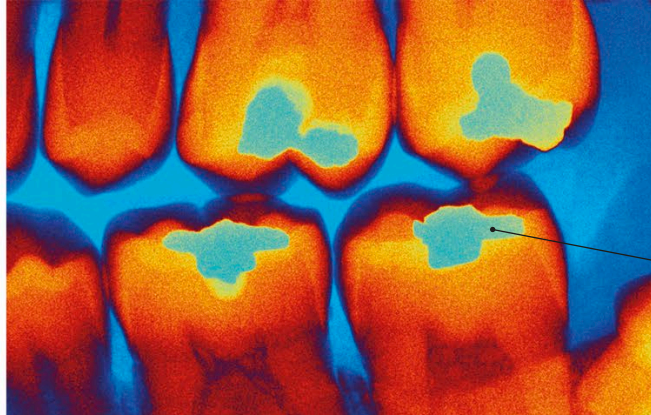




Colored X-ray
of teeth

Amalgam filling

FILLING IN THE HOLES

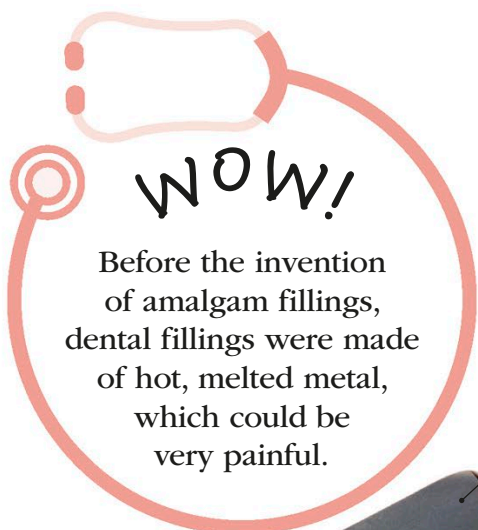
In the 1820s, several dentists, including Auguste Taveau in France and Thomas Bell in the UK, started using a new dental filling called amalgam. Made from silver and mercury, it could be easily molded to fit a cavity in a tooth. In the 20th century, more natural-looking white fillings made of powdered glass, ceramic, and plastic were developed.



Modern dental braces

DENTAL BRACES

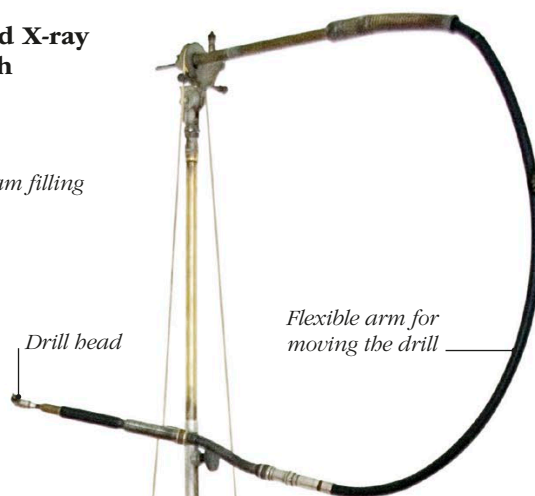
The first wire braces for pulling the teeth into position were made by the French dentist Christophe-François Delabarre in 1819. Various improvements have been made since then, including the addition of elastic bands in the 1840s and the development of dental adhesives in the 1970s to keep the braces in place.



WOW!
Before the invention
of amalgam fillings,
dental fillings were made
of hot, melted metal,
which could be
very painful.



Adjustable
footrest



Drill head

Flexible arm for
moving the drill

REMOVING DECAY

Dental drills are mainly used for removing decayed parts of teeth. They underwent a major evolution from the mid-19th century onward, beginning with the introduction of the clockwork drill in 1864, followed by the electric drill in 1875. But these were surpassed by the compressed air-powered turbine drill in 1949. Invented by the New Zealand dentist John Patrick Walsh, its drill head could spin at the rate of 400,000 rpm (rotations per minute)—compared to just 3,000 rpm for an electric drill.

◀ SPEEDING IT UP

Invented by the American dentist James Beall Morrison in 1871, the foot-powered "treadle" drill sped up dental treatment.

Rope and pulley system
operates the drill head.

Foot pedal
is pushed up
and down to
turn the drill.

**Treadle drill,
1871**

